

Arman Goshtasbi

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Overview

- Skilled in CAD modeling and mechanical design using SolidWorks and Rhino/Grasshopper
- Proficient in additive manufacturing, including direct ink writing (DIW), fused filament fabrication (FFF), and lost-wax casting
- Experienced in machine design and development, particularly for custom FFF and DIW systems, using custom CNC electronics
- Strong background in experimental design, prototyping, and comprehensive data collection
- Advanced skills in data analysis and processing, with a solid programming foundation in MATLAB and Python
- Hands-on experience with fabrication techniques, with a focus on soft robotics

Education

Ph.D. in Biorobotics	2023 – 2026
<i>University of Southern Denmark</i>	Odense, Denmark
M.S. in Mechanical Engineering (Robotics Specialization)	2020 – 2023
<i>University of Twente</i>	Enschede, The Netherlands
B.Sc. in Mechanical Engineering	2016 – 2020
<i>University of Tehran</i>	Tehran, Iran

Skills

Programming languages	MATLAB, Python, Arduino IDE (proficient)
CAD software	SolidWorks, Rhino/Grasshopper
Simulation software	COMSOL Multiphysics
Additive manufacturing	Machine development, Tool-path design, G-code generation
Others	Adobe Premier, Adobe Illustrator, L ^A T _E X, SIMATIC S7

Research Experience

Visiting PhD Researcher	Mar 2025 – Jun 2025
<i>ETH Zurich, Complex materials</i>	
• Developed silicone-based conductive ink compatible with commercially used soft robotic materials • Characterized rheological properties of the ink to identify optimal shear-thinning behavior for precision DIW. • Fabricated sensor-integrated soft robots using a custom-built 5-axis DIW printing platform.	
Graduate Research Assistant (Ph.D.)	Jun 2023 – present
<i>University of Southern Denmark, Soft Robotics Lab</i>	
• Researched and developed monolithic soft robotic actuator fabrication using wax 3D printing, integrating actuator design, prototyping, and post-processing. • Developed high-resolution 1D and 2D soft tactile sensors using fluid-filled channels and external pressure sensing. • Designed and implemented an experimental platform to characterize sensor behavior under various mechanical stimuli. • Conducted extensive data acquisition and analysis of pressure signals across multiple fluid media. • Trained and validated a Fuzzy Clustering model to estimate touch position and force based on pressure profiles. • Demonstrated real-time human–robot interaction using the developed sensor integrated with a UR robotic arm. • Designed a custom, open-source CNC control box to enable versatile multi-axis motion for applications including DIW and textile-based soft actuator fabrication. • Developed and integrated a multi-tool CNC platform combining ultrasonic welding and oscillating cutting for automated manufacturing of inflatable textile actuators.	

- Tested a broad range of textile materials and actuator designs, demonstrating its versatility.

Graduate Research Assistant (M.S.)

Jan 2022 – May 2023

University of Twente, Soft Robotics Lab

- Developed an automated continuous fiber embedding technique using a modified FDM printer to enable complex 3D weaving for soft robotics applications.
- Designed and fabricated fiber-reinforced hybrid robotic joints combining soft and rigid components to improve mechanical integration and load capacity.
- Invented a bioinspired suction cup with granular jamming for stiffness modulation, enabling passive adaptation and strong attachment to angular surfaces.
- Introduced an FDM-based underextrusion method for bonding soft and rigid materials, outperforming conventional adhesives in shear and peel strength.
- Validated the underextrusion method through mechanical testing, pressure endurance trials, and soft robotic demonstrators.

Mechanical Engineering Intern

Sep 2021 – Jan 2022

Eindhoven University of Technology

- Solved the inverse kinematics of an extensible multi-segment soft pneumatic arm using Piecewise Constant Curvature and rigid-link approximation.
- Applied the inverse kinematics solution for controlling tip angle and optimizing pressure.
- Implemented and visualized the kinematic model in a virtual reality (VR) environment using Unity to simulate soft robot bending behavior.

Graduate Research Assistant

Mar 2021 – Jun 2021

University of Twente, Surgical Robotics Lab

- Modeled and analyzed a new electromagnetic end-effector using finite element methods for the Advanced Robotic for Magnetic Manipulation (ARMM) system in COMSOL.
- Designed and modeled a liquid-based cooling system for the end-effector using SolidWorks and COMSOL.

Undergraduate Research Assistant

Oct 2018 – Dec 2019

University of Tehran in collaboration with University of Twente

- Designed a high-strength aluminium solid-state additive manufacturing tool using SolidWorks.
- Analyzed the heat transfer of the tool cooling system using COMSOL.
- Studied the effect of tool rotation speed on the grain size of the clad layer.

Publication (Google Scholar)

Journal Papers

- **Goshtasbi, A et al.**, ‘Tactile Perception through Fluid–Solid Interaction’, Nature Communications, 2026 (Link)
- **Goshtasbi, A.**, Grignaffini, L., Sadeghi, A., “From Problem to Solution: Bio-inspired 3D Printing for Bonding Soft and Rigid Materials via Underextrusion”, Scientific Report, 2025 (Link)
- **Goshtasbi, A.**, Sadeghi, A., “A Bioinspired Stiffness Tunable Sucker for Passive Adaptation and Firm Attachment to Angular Substrates”, Frontiers in Robotics and AI, 2023, (Link)
- Keyvanara, M., **Goshtasbi, A.**, Kuling, I.A., “A Geometric Approach Towards Inverse Kinematics of Soft Extensible Pneumatic Actuators Intended for Trajectory Tracking”, Sensors, 2023 (Link)

Selected Conference Papers & Presentations

- **Goshtasbi, A et al.**, “Weld n’Cut: Automated fabrication of inflatable fabric actuators”, IEEE 8th International Conference on Soft Robotics (RoboSoft), 2025 (Link)
- Parvaresh, A., **Goshtasbi, A et al.**, Tirado, J., Rafsanjani, A. “Proprioceptive Origami Manipulator”, IEEE 8th International Conference on Soft Robotics (RoboSoft), 2025 (Link)

• **Goshtasbi, A et al.**, “AI-infused soft fluidic tactile sensing”, IEEE 7th International Conference on Soft Robotics (RoboSoft), 2024 (Link)

HONOURS & AWARDS

Fondation Idella travel grant	May 2024
Thomas B. Thrige Foundation travel grant	March 2025
Knud Højgaards Fond travel grant	June 2025
Certificate	

Basic Programming of PLCS-PLC211 August 2019

- Festo Company
- Basic command set for SIMATIC S7.
 - PLC timers and counters.
 - Programming using organization block.

References

Ahmad Rafsanjani Professor SDU Biorobotics University of Southern Denmark +4565503961 ahra@mmmi.sdu.dk	Ali Sadeghi Assistant Professor Department of Biomechanical Engineering University of Twente +31-5-34899066 a.sadeghi@utwente.nl
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